# Hands-on Lab: Data Generation and Cleansing

## Estimated time: 30 minutes

## Learning objectives

After completing this lab, you will be able to:

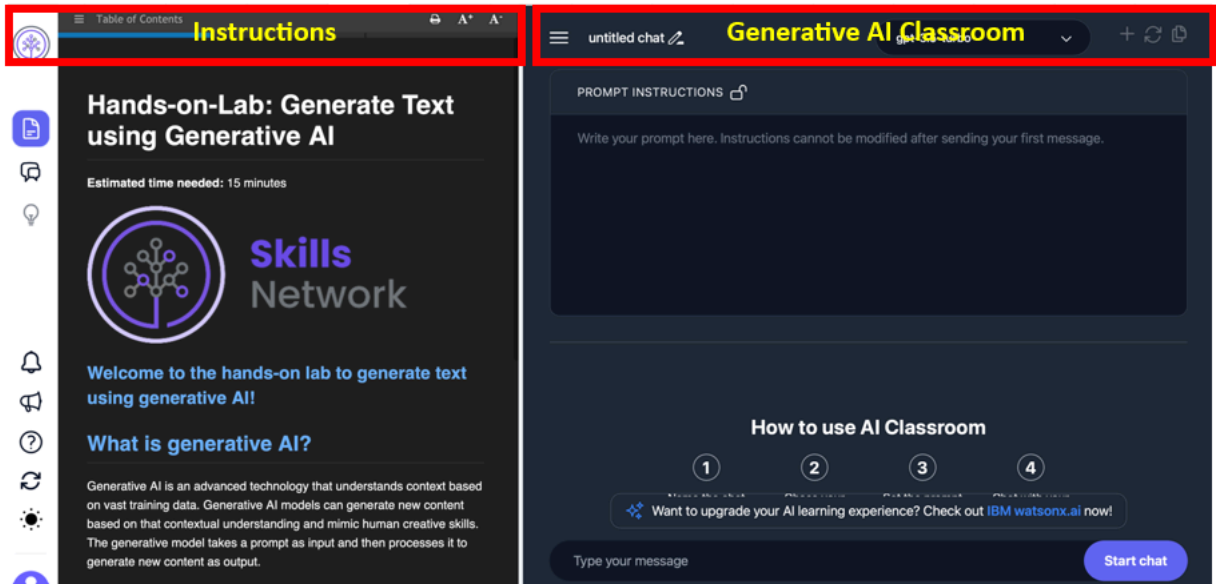
- Generate synthetic data using generative AI prompts.
- Perform basic data cleansing tasks to prepare your data set for analysis.

Welcome to the lab on data generation and cleansing!

# About generative AI classroom lab

The generative AI classroom is a tool where you can write and compare your prompts to generate desired text with real-time chat responses. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

You will have both the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen. You can interact with the language model using the message and chat fields.

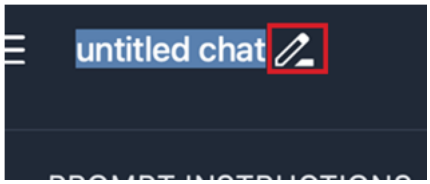



### Set up the AI classroom

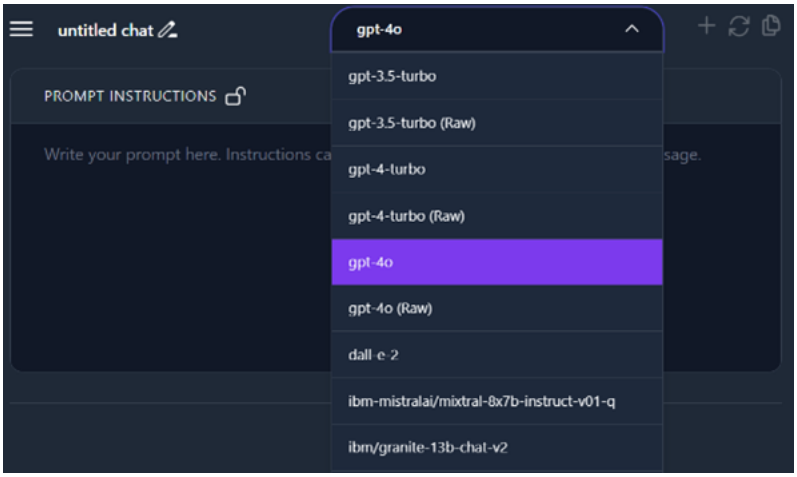
To work on an exercise in the AI classroom, you must first set up your AI classroom for a better learning experience.

Before beginning the exercise, you must:

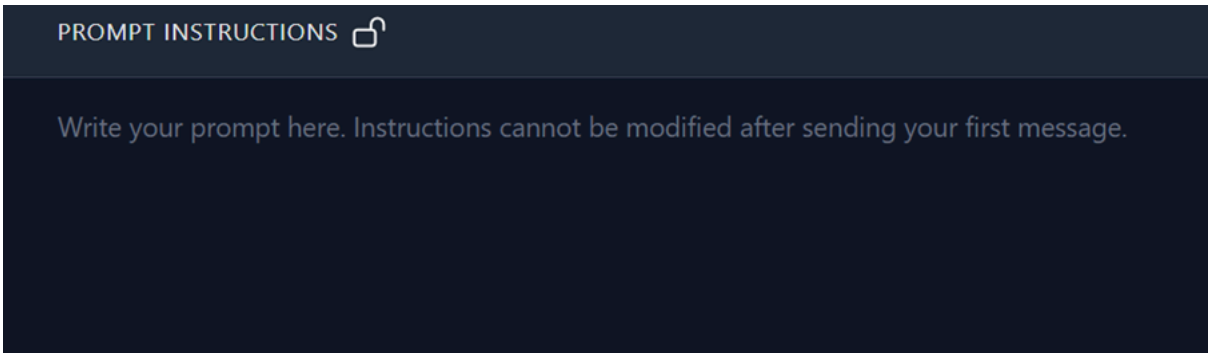
1. **Name the chat:** Use the pencil icon available on the top-left corner of the right pane to name the chat.




2. **Choose the model:** Use the dropdown option from the top-right corner of the right pane for the text generation exercise. In this lab, we will work with the **gpt-4o** model.




3. **Give prompt instructions:** Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.




4. **Type your message:** Use the text box available at the bottom of the page to write the prompts and converse with the chat system.

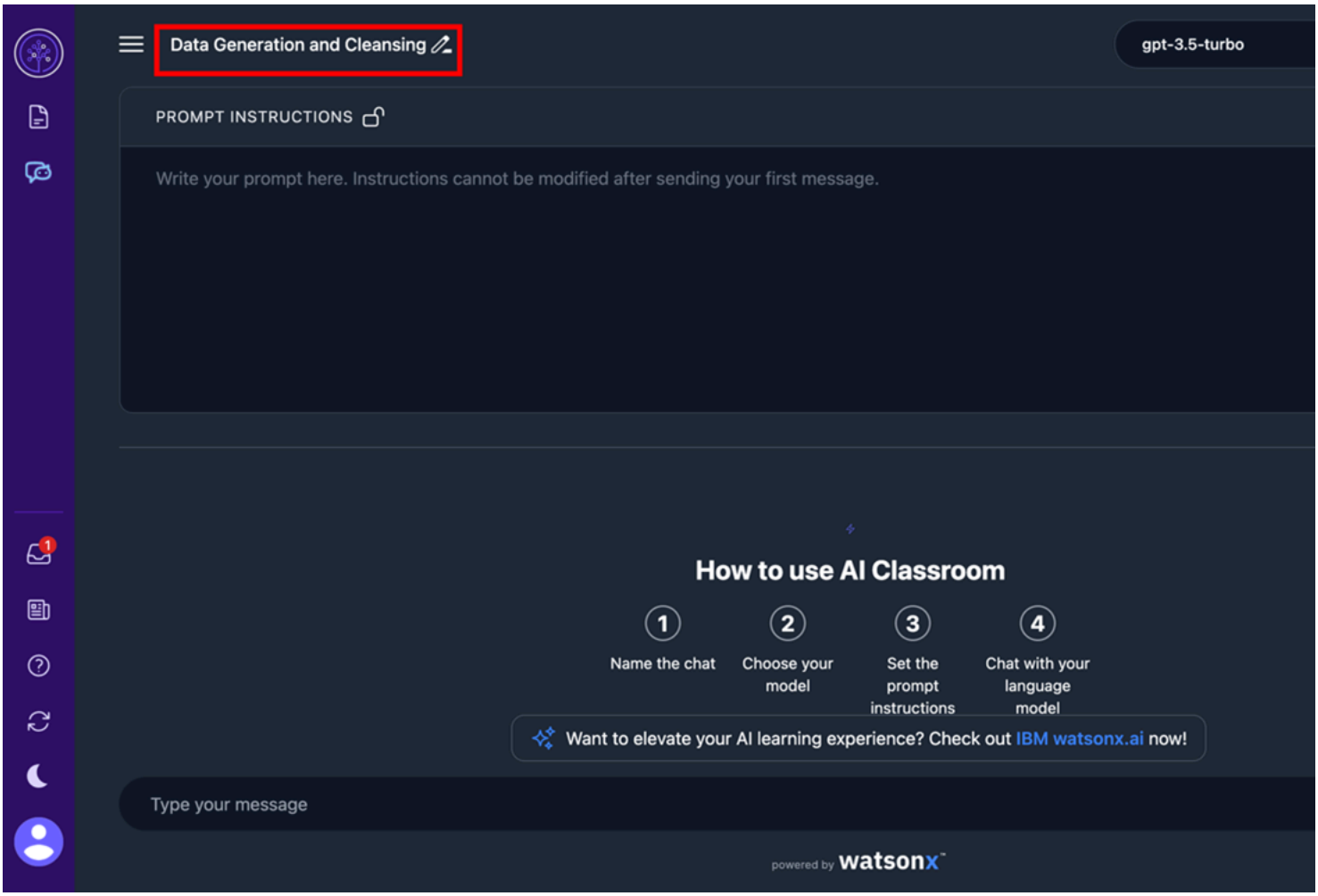
**Note: The prompts used in this lab are for your reference only. You can create your own prompts and generate responses using generative AI. You may receive different responses even though you've used the same prompt from this lab.**

# Exercise: Generate data synthetically and perform a basic cleansing task on the data

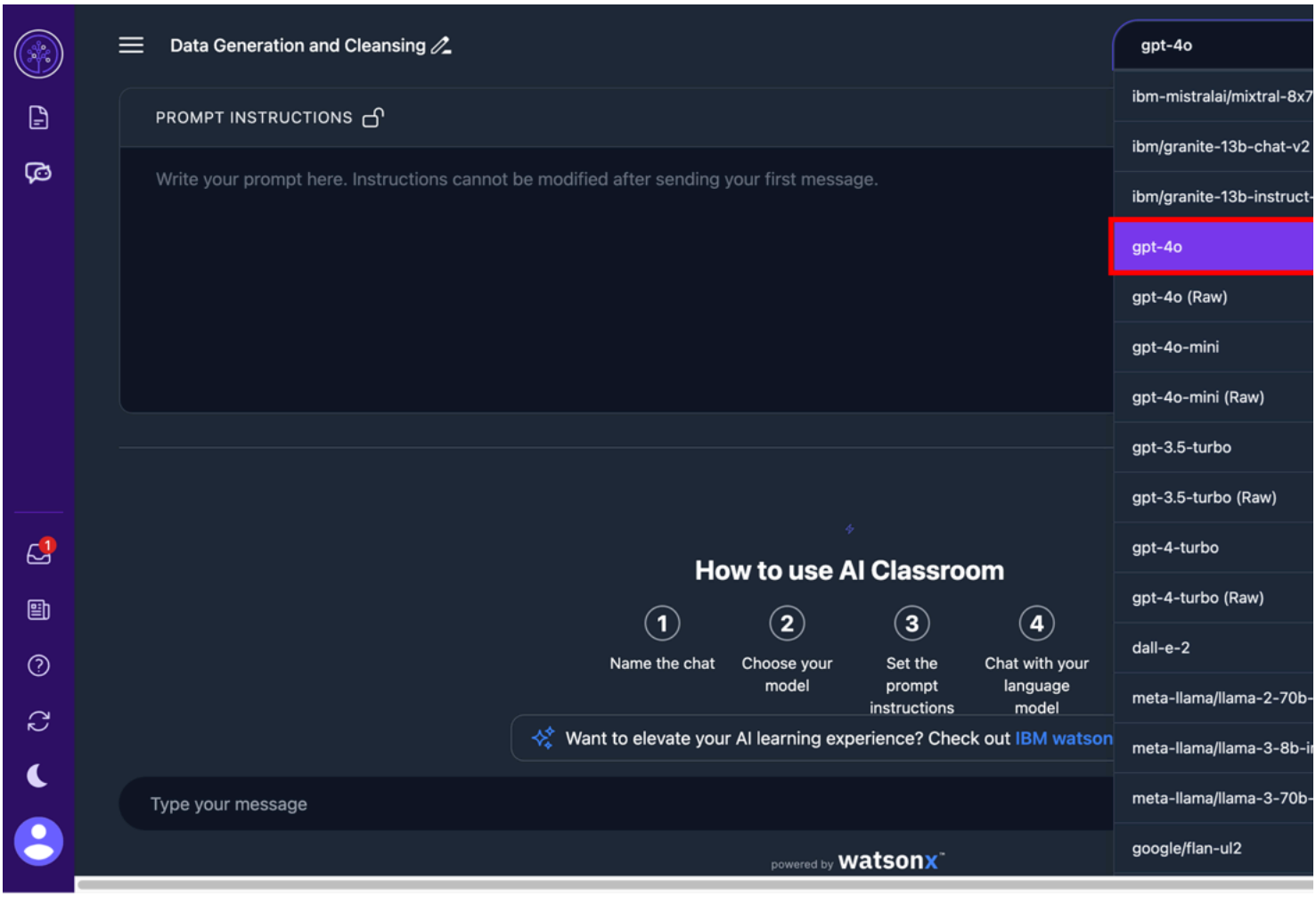
In this exercise, you will draft prompts in the generative AI classroom chatbot to generate synthetic data and perform basic cleansing tasks like handling missing values, removing duplicates, and standardizing data.
Let's begin.
You are an analyst for a departmental store chain. At times, you need to augment their existing data sets with synthetically generated data to get statistically significant results. Then, you also need to perform basic cleansing operations on the data set to prepare the data for the analysis stage.

### Task 1: Generate synthetic data

This is the interface you'll see in the beginning. Before you begin with the task, name the chat. Call it **"Data Generation and Cleansing."**



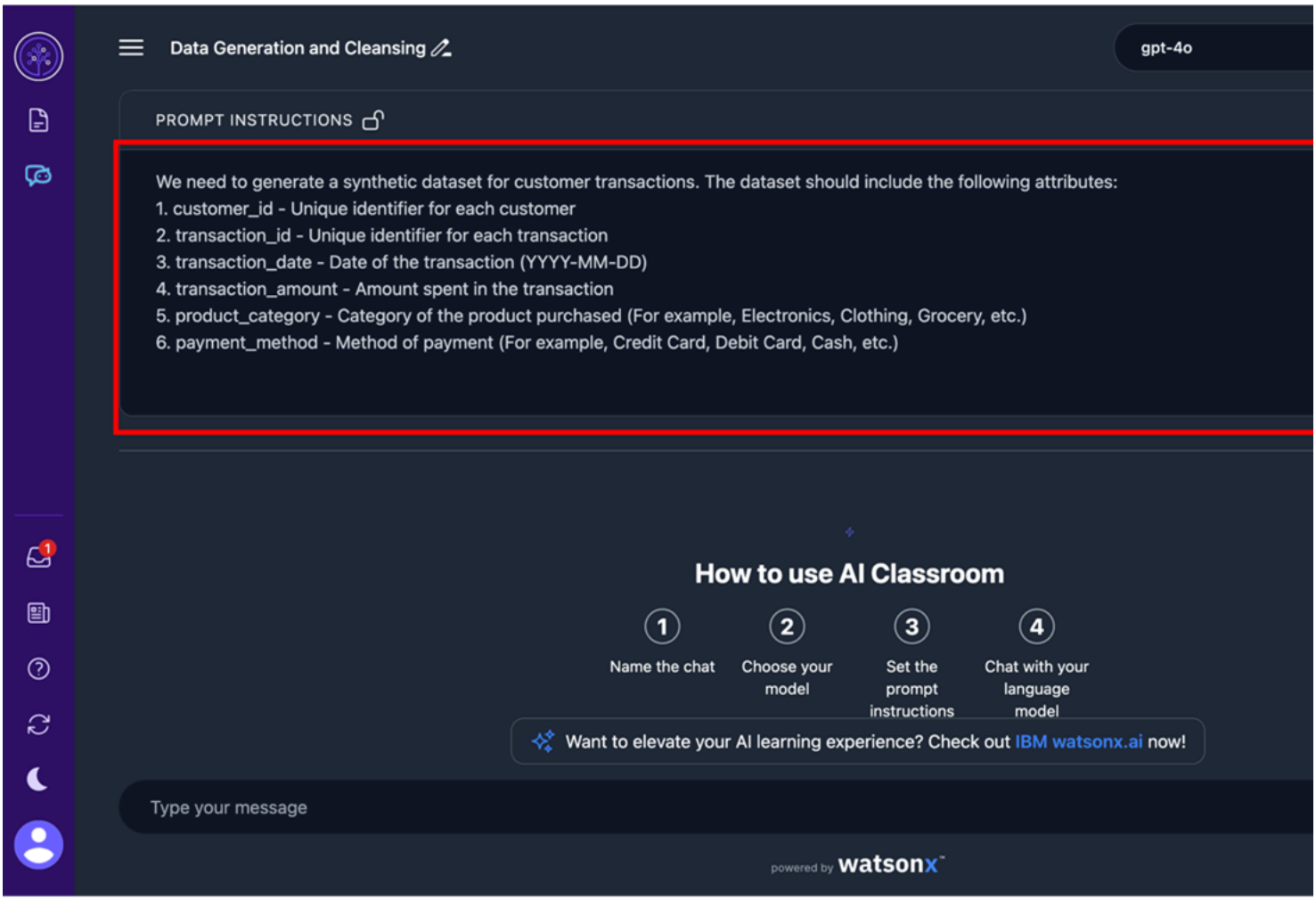

Select a model **gpt-4o** from the drop-down list.




**Step 1: Setting the context**

Provide the generative AI model with the necessary context to generate the synthetic data set. Paste the following instruction in the Prompt Instructions field:

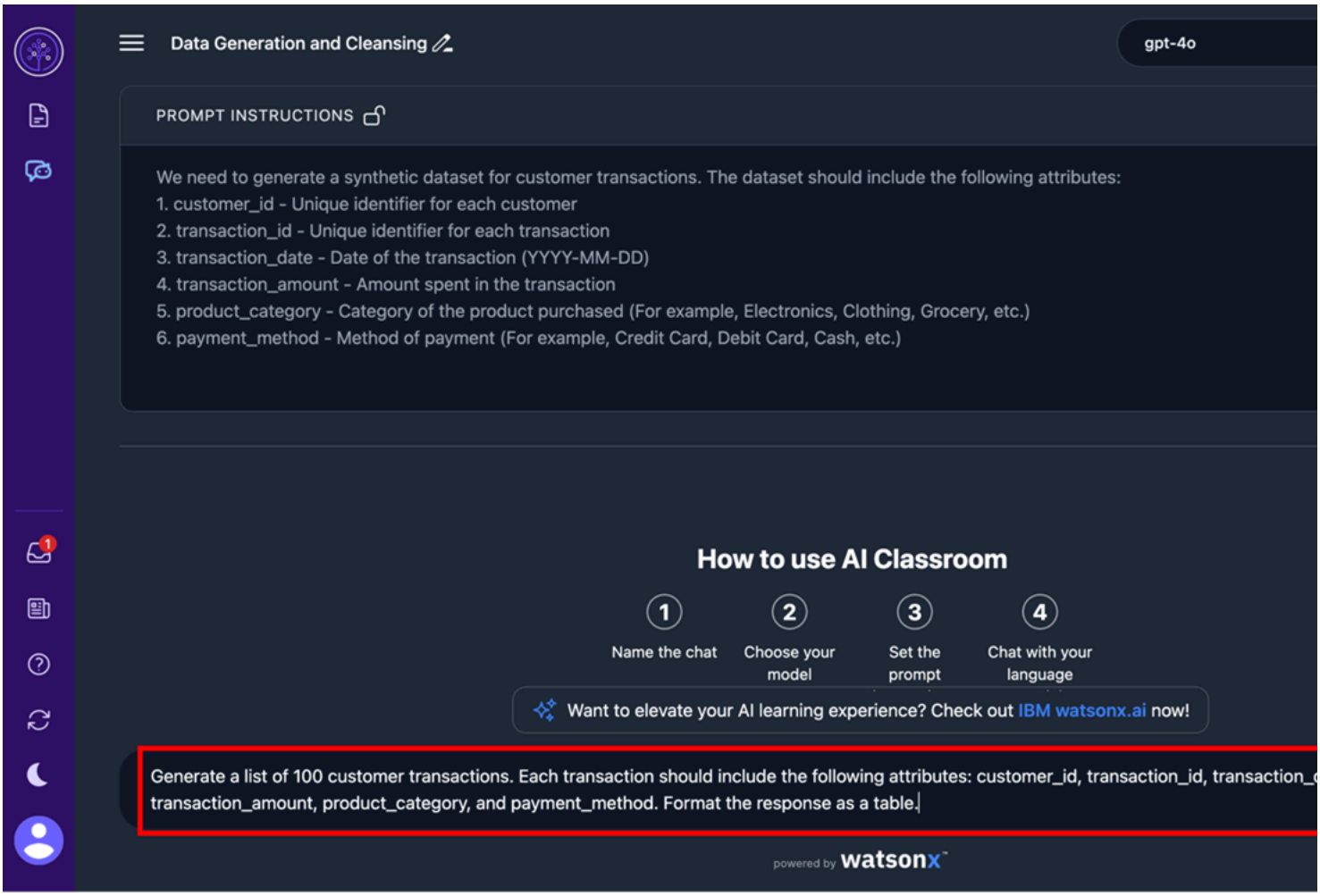
```
We need to generate a synthetic dataset for customer transactions. The dataset should include the following attributes:
1. customer_id - Unique identifier for each customer
2. transaction_id - Unique identifier for each transaction
3. transaction_date - Date of the transaction (YYYY-MM-DD)
4. transaction_amount - Amount spent in the transaction
5. product_category - Category of the product purchased (For example, Electronics, Clothing, Grocery, etc.)
6. payment_method - Method of payment (For example, Credit Card, Debit
Card, Cash, etc.)
```



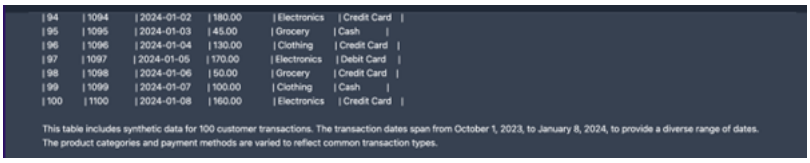

**Step 2: Generating the data in batches**

Break down the request into smaller parts, asking for a manageable number of transactions in each prompt.
Enter the following prompt and select the **Start Chat** button.

```
Generate a list of 100 customer transactions. Each transaction should include the following attributes: customer_id, transaction_id, transaction_date
```




AI generates 100 customer transactions with the specified columns. Screenshots showing the first few and last few rows are shown below.




You know how you can generate synthetic data. Now, let's cleanse the data set to prepare it for data analysis.

### Task 2: Cleanse the data

You'll notice that the data you have generated is complete, has no discrepancies, and is ready for analysis. To perform the cleansing tasks, you need a data set with inconsistencies within it. Let's generate one such data set.

Enter the following prompt and select the **Send** button.

```
Generate a list of 50 customer transactions. Each transaction should include the following attributes: customer_id, transaction_id, transaction_date,
Please include around three instances of missing values in transaction amount and three instances in the product category and four instances of dupli
```

"Generate a list of 50 customer transactions. Each transaction should include the following attributes: customer_id, transaction_id, transaction_date, transaction_amount, product_category, and payment_method. Format the response as a table.
Please include around three instances of missing values in transaction amount and three instances in the product category and four instances of duplicate entries. The product_category and payment_category must have inconsistent casing."

Send

This will generate a data set with the specified inconsistencies for you. Screenshots showing the first few and last few rows are shown below.

Sure! Below is a table of 50 customer transactions with the specified attributes, including instances of missing values, inconsistent casing for product categories and payment methods, and duplicate entries.

| customer_id | transaction_id | transaction_date | transaction_amount | product_category | payment_method |
|---|---|---|---|---|---|
| C001 | T001 | 2023-10-01 | 150.75 | electronics | Credit Card |
| C002 | T002 | 2023-10-01 | 89.99 | Clothing | debit card |
| C003 | T003 | 2023-10-02 | 45.50 | Grocery | Cash |
| C004 | T004 | 2023-10-02 | | Electronics | Credit Card |
| C005 | T005 | 2023-10-03 | 120.00 | clothing | Debit Card |
| C006 | T006 | 2023-10-03 | 75.25 | Grocery | cash |
| C007 | T007 | 2023-10-04 | 200.00 | electronics | |

Regenerate response

| | | | | | |
|---|---|---|---|---|---|
| C044 | T044 | 2023-10-22 | 90.99 | clothing | debit card |
| C045 | T045 | 2023-10-23 | 50.00 | Grocery | Cash |
| C046 | T046 | 2023-10-23 | 210.75 | electronics | Credit Card |
| C047 | T047 | 2023-10-24 | 85.00 | Clothing | Debit Card |
| C048 | T048 | 2023-10-24 | 45.50 | grocery | cash |
| C049 | T049 | 2023-10-25 | 180.00 | Electronics | Credit Card |
| C050 | T050 | 2023-10-25 | 95.99 | clothing | debit card |

Note: Duplicate entries are for customer_id C001, transaction_id T001. The missing values are present in transaction_amount and product_category as specified. Inconsistent casing is applied to product_category and payment_method.

Now we have a data set on which we can perform the data cleansing steps.

**Step 1: Handling missing data**

To handle the missing data, enter the prompt below.

```
Identify and fill any missing values in the dataset. For numeric attributes, use the mean to fill missing values. For categorical attributes, use the
```

Identify and fill any missing values in the dataset. For numeric attributes, use the mean to fill missing values. For categorical attributes, use the mode to fill missing values. Provide the updated dataset.

Send

The AI generates an output, as shown below. Here are the screenshots showing the beginning of the output and the last few lines.

Sure! Below is the updated dataset with missing values filled. For numeric attributes, the mean has been used, and for categorical attributes, the mode has been used.

First, let's calculate the mean for `transaction_amount` and the mode for `product_category`:

- Mean of `transaction_amount` (excluding missing values):
$$\text{Mean} = \frac{150.75 + 89.99 + 45.50 + 120.00 + 75.25 + 200.00 + 60.99 + 30.00 + 99.00 + 55.50 + 180.00 + 75.99 + 40.00 + 220.75 + 70.00 + 35.50 + 150.00 + 60.00 + 210.50 + 95.99 + 50.00 + 175.00 + 85.75 + 40.50 + 190.00 + 70.00 + 45.00 + 130.00 + 65.00 + 200.99 + 85.00 + 50.50 + 170.00 + 80.75 + 35.00 + 180.00 + 75.00 + 40.50 + 200.00 + 90.99 + 50.00 + 210.75 + 85.00 + 45.50 + 180.00 + 95.99}{46}$$
$$\approx 108.25$$

Regenerate response

- Mode of `product_category`:
$$\text{Mode} = \text{Electronics}$$

Now, we can fill in the missing values:

| customer_id | transaction_id | transaction_date | transaction_amount | product_category | payment_method |
|---|---|---|---|---|---|
| C001 | T001 | 2023-10-01 | 150.75 | electronics | Credit Card |
| C002 | T002 | 2023-10-01 | 89.99 | Clothing | d |
| C003 | T003 | 2023-10-02 | 45.50 | Grocery | ( |

Regenerate response

| | | | | | |
|---|---|---|---|---|---|
| C044 | T044 | 2023-10-22 | 90.99 | clothing | debit card |
| C045 | T045 | 2023-10-23 | 50.00 | Grocery | Cash |
| C046 | T046 | 2023-10-23 | 210.75 | electronics | Credit Card |
| C047 | T047 | 2023-10-24 | 85.00 | Clothing | Debit Card |
| C048 | T048 | 2023-10-24 | 45.50 | grocery | cash |
| C049 | T049 | 2023-10-25 | 180.00 | Electronics | Credit Card |
| C050 | T050 | 2023-10-25 | 95.99 | clothing | debit card |

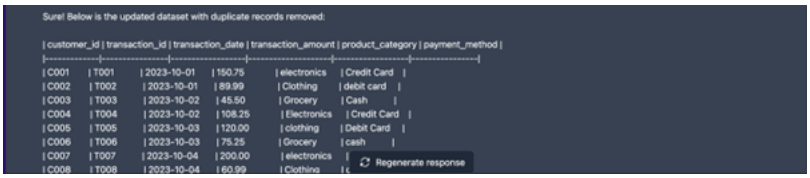
Note: Duplicate entries are for customer_id C001, transaction_id T001. The missing values are filled as specified. Inconsistent casing is maintained for product_category and payment_method.

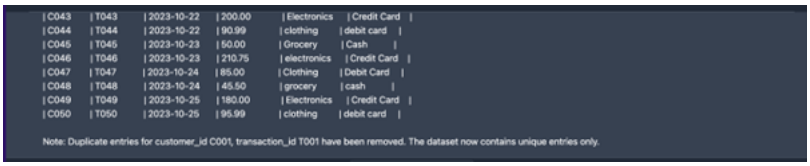
**Step 2: Removing duplicates**

Next, let's remove the duplicates. Enter the following prompt and select the **Send** button.

```
Identify and remove any duplicate records in the dataset. Provide the updated dataset.
```

The AI generates an output, as shown below. Here are the screenshots showing the beginning of the output and the last few lines.

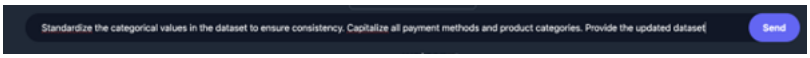


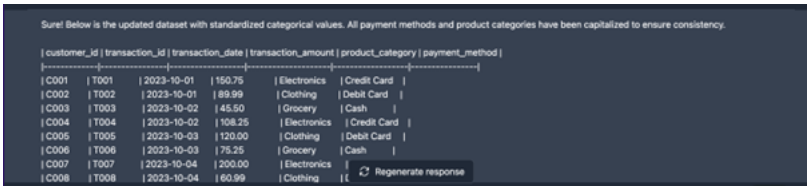

**Step 3: Standardizing categorical values**

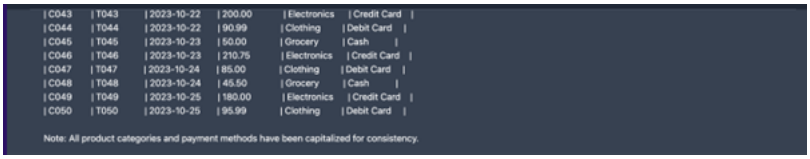
Further, let's standardize the casing for the product category and payment method columns. Enter the following prompt and select the **Send** button.

```
Standardize the categorical values in the dataset to ensure consistency. Capitalize all payment methods and product categories. Provide the updated c
```




The AI generates an output, as shown below. Here are the screenshots showing the beginning of the output and the last few lines.




Now that you have performed three basic data cleansing tasks, you can practice data querying and cleansing using the following prompts.

# Try it yourself

Try the following prompts yourself.

**Prompt 1: Identify outliers**
Identify any outliers in the 'transaction_amount' attribute using statistical methods such as the IQR (Interquartile Range). Provide suggestions on how to handle these outliers.

**Prompt 2: Normalize numeric attributes**
Normalize the 'transaction_amount' attribute to have a mean of 0 and a standard deviation of 1. Provide the updated data set.

**Prompt 3: Correct invalid categorical values**
Validate the 'product_category' values against the predefined list ["Electronics", "Clothing", "Grocery", "Home & Kitchen", "Books"]. Correct any invalid entries and provide the updated data set.

**Prompt 4: Convert date formats**
Ensure that all 'transaction_date' values are in the standard format YYYY-MM-DD. Convert any dates that are not in this format and provide the updated data set.

**Prompt 5: Segment customers by spending**
Create a new attribute 'spending segment' that segments customers into "Low", "Medium", and "High" based on their total spending. Define the thresholds for each segment and provide the updated data set.

**Prompt 6: Analyze age distribution**
Generate a summary of the age distribution of customers, including minimum, maximum, and average age.

**Prompt 7: Gender analysis**
Count the number of male and female customers in the data set and provide the result.

## Summary

Congratulations on completing the hands-on lab on data generation and cleansing.

In this lab, you've explored how to use generative AI to generate synthetic data and perform data cleansing tasks like handling missing values, removing duplicates, and standardizing data.